

UNITED STATES PATENT AND TRADEMARK OFFICE

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Title: PROCESS FOR THE PREPARATION OF METHYL 4-HYDROXYBENZOATE AND INTERMEDIATES THEREOF

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ABSTRACT

Disclosed is an improved process for preparing methyl 4-hydroxybenzoate (paraben) from 4-hydroxybenzoic acid by ac

BACKGROUND OF THE INVENTION

Parabens are widely used preservatives in pharmaceutical formulations. Known processes employ sulfuric acid or p-tol

SUMMARY OF THE INVENTION

The present invention provides a process comprising: esterifying 4-hydroxybenzoic acid with methanol in the presence

DETAILED DESCRIPTION

In a preferred embodiment, 4-hydroxybenzoic acid (CAS 99-96-7) is suspended in methanol. Concentrated sulfuric acid

EXAMPLE 1

Preparation of methyl 4-hydroxybenzoate. 4-hydroxybenzoic acid (13.8 g, 100 mmol) and methanol (50 mL) were charged

EXAMPLE 2

Preparation of sodium 4-hydroxybenzoate. Methyl 4-hydroxybenzoate (7.6 g, 50 mmol) was dissolved in ethanol (30 mL).

INDUSTRIAL MANUFACTURING PROCESS

At industrial scale, the esterification is carried out in a glass-lined reactor of 1000 L. The reactor is charged with 4-hydroxybenzoic acid, methanol and a strong acid.

WHAT IS CLAIMED IS:

1. A process for preparing methyl 4-hydroxybenzoate comprising esterifying 4-hydroxybenzoic acid with methanol in the presence of a strong acid.
2. The process of claim 1, wherein the strong acid is sulfuric acid.
3. The process of claim 1, further comprising purifying by crystallization.